



RATIOS



OBJECTIVES

- 1 Define ratios
- 2 Types of ratios
- 3 Visual models
- 4 Solving ratio word problems

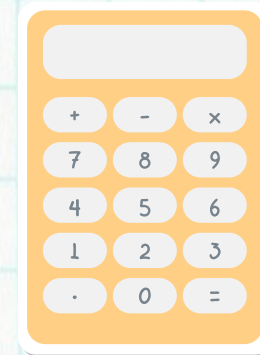
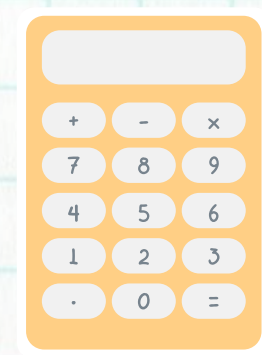
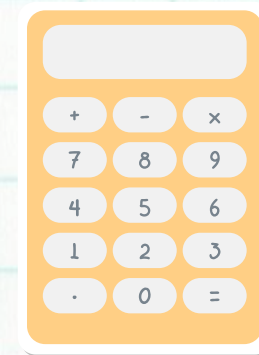
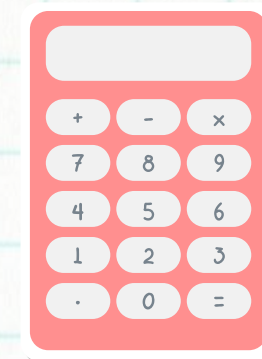


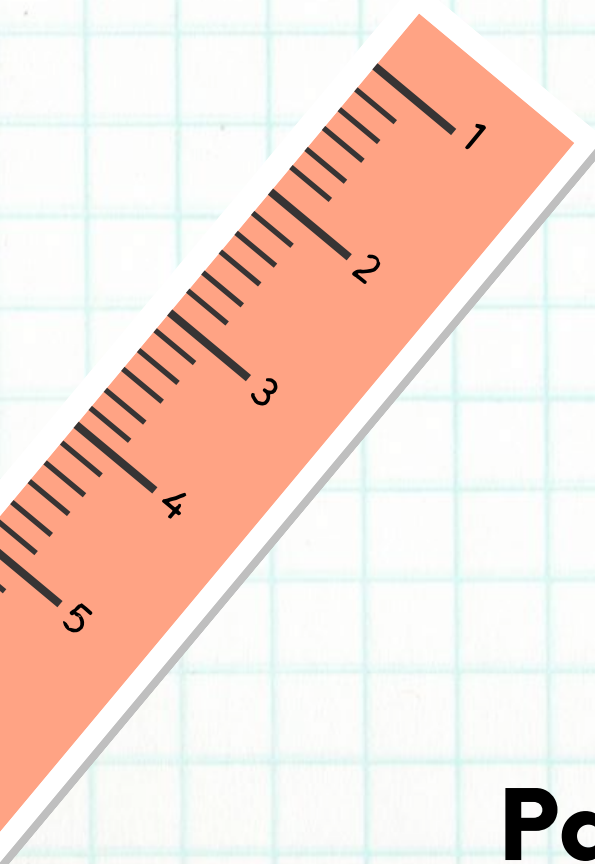
INTRODUCTION

A ratio tells us how much of one thing there is compared to another thing.

FOR EXAMPLE

For every two red calculators we have three yellow calculators.





TYPES OF RATIOS

1

Part-to-Part

Compares the number of one part to another part of the whole.

2

Part-to-Whole

Compares a selected number of parts to the total number of all parts in the whole.

3

Whole-to-Part

Compares the total number of all parts in the whole to a selected number parts.





WAYS TO SHOW RATIOS

The same ratio can be written in three different ways.

1

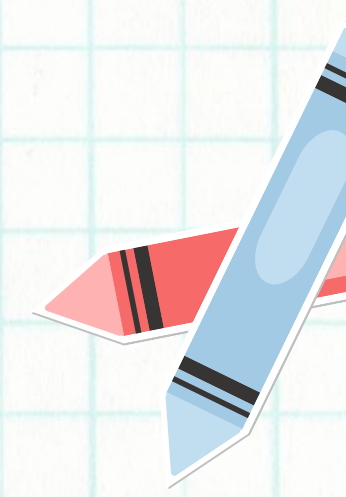
four to six

2

4:6

3

$\frac{4}{6}$



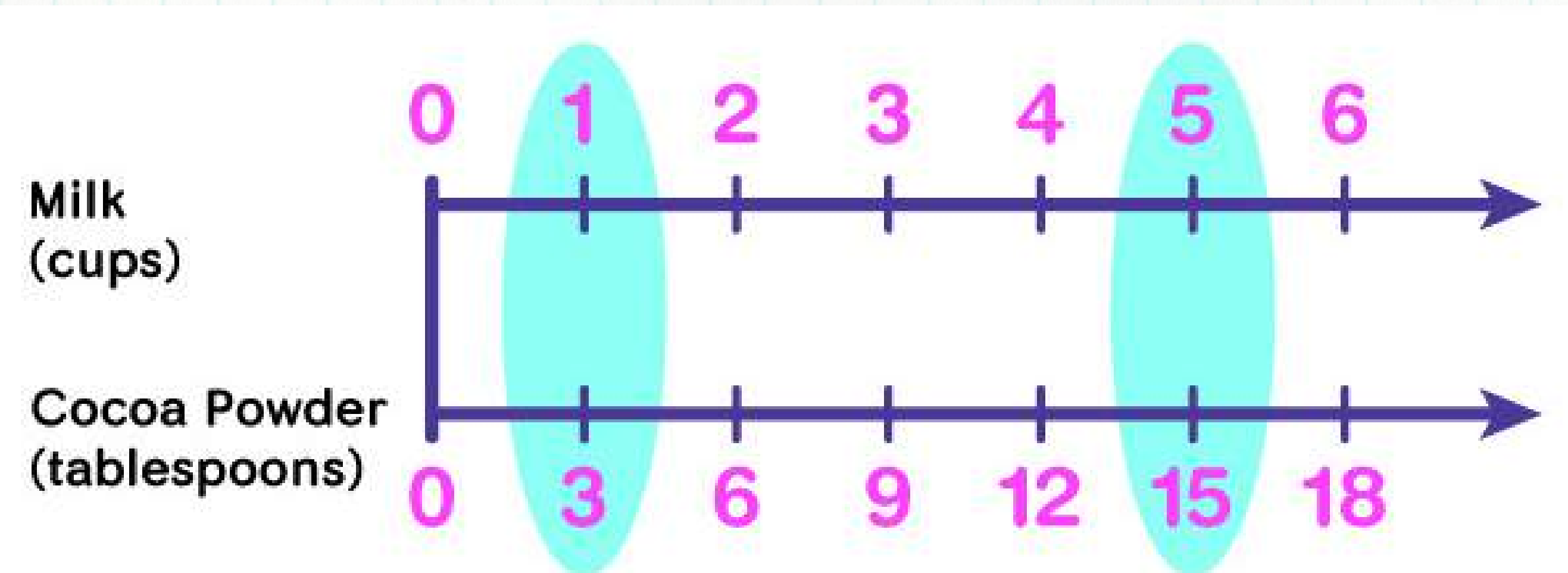
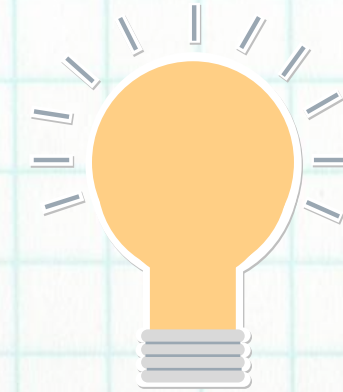
RATIO TABLES

a	b
1	4
2	8
3	12
6	24
9	36

- Helps us understand equalvalent ratios
- Shows how 2 related quantities grow or think by multiplying or dividing both sides by the same number

DOUBLE NUMBER LINES

- Used to compare quantities
- Shows growth of 2 quantities



SIMPLIFYING RATIOS

Just like fractions

Use GCF

E.g., $12:16 =$ Divide
both by 4 $= 3:4$



EQUIVALENT RATIOS

Multiply or divide both terms by
the **same number**.

$$2:3 =$$

$$\times 2 = 3:6$$

$$\times 3 = 6:9$$

$$\times 5 = 10:15$$



SOLVING RATIOS

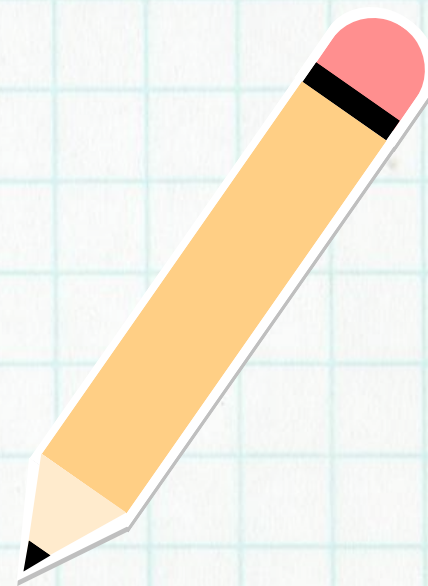
1

Identify how many of each item is shown.



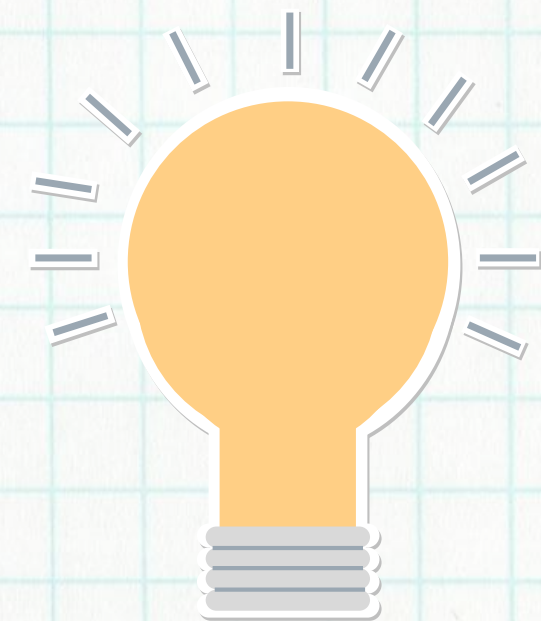
2

Write your ratio.



3

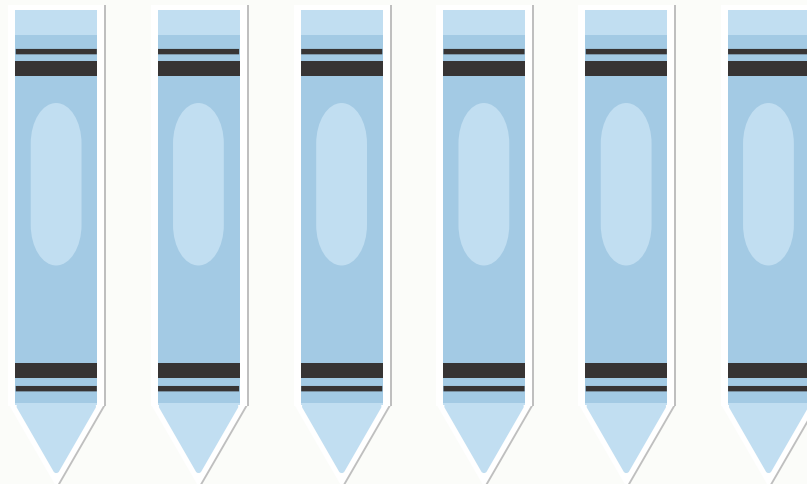
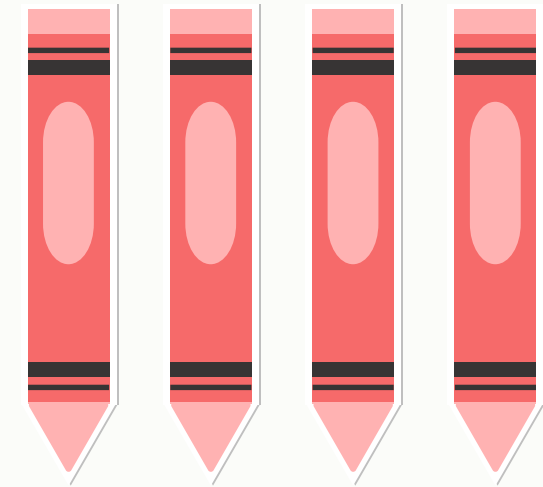
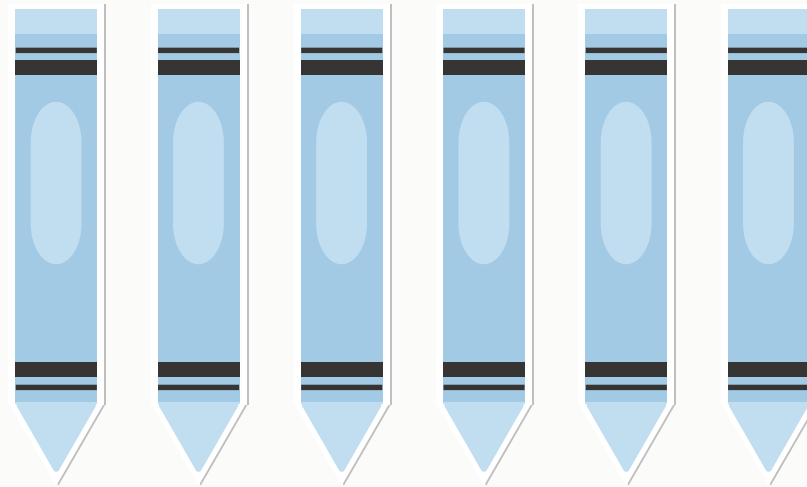
Simplify, if possible.



SOLVING RATIOS

1

Identify how many of
each item is shown.



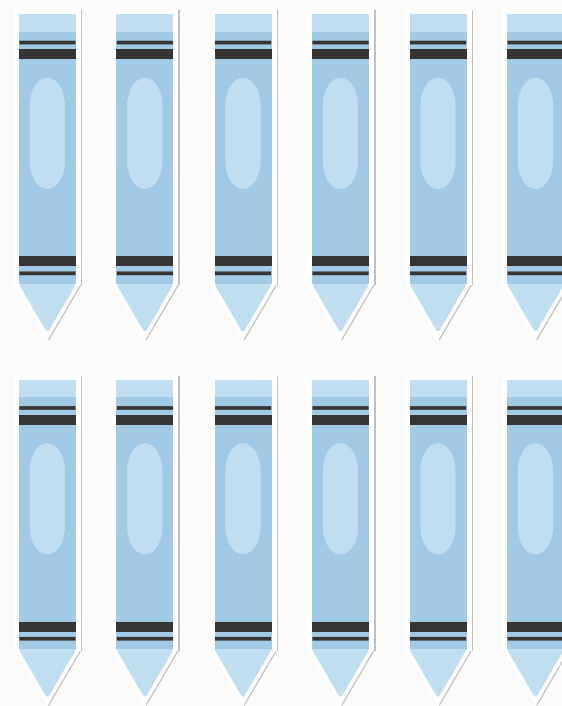
SOLVING RATIOS

2

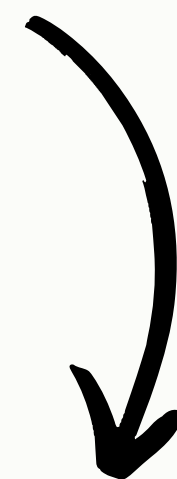
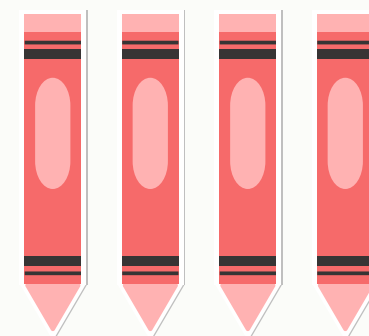
Write your ratio.

Write the ratio of
blue crayons to red
crayons in simplest
form.

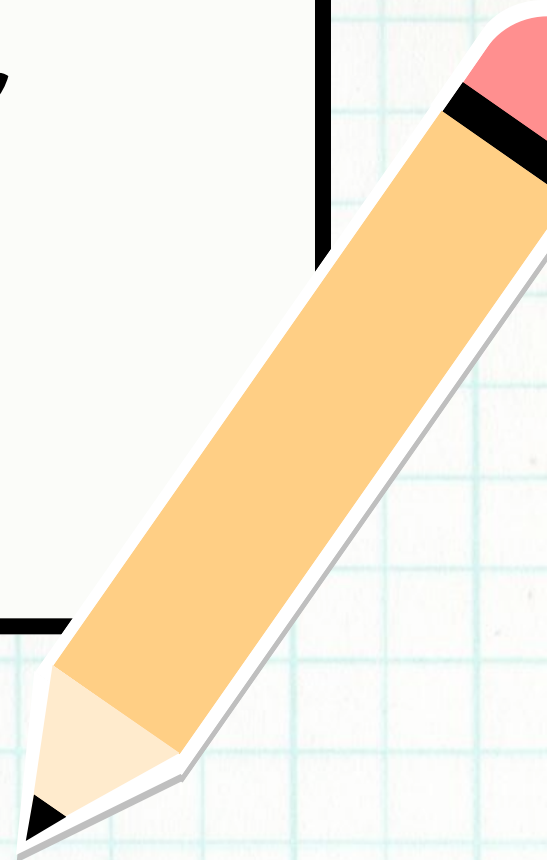
12



4



12:4

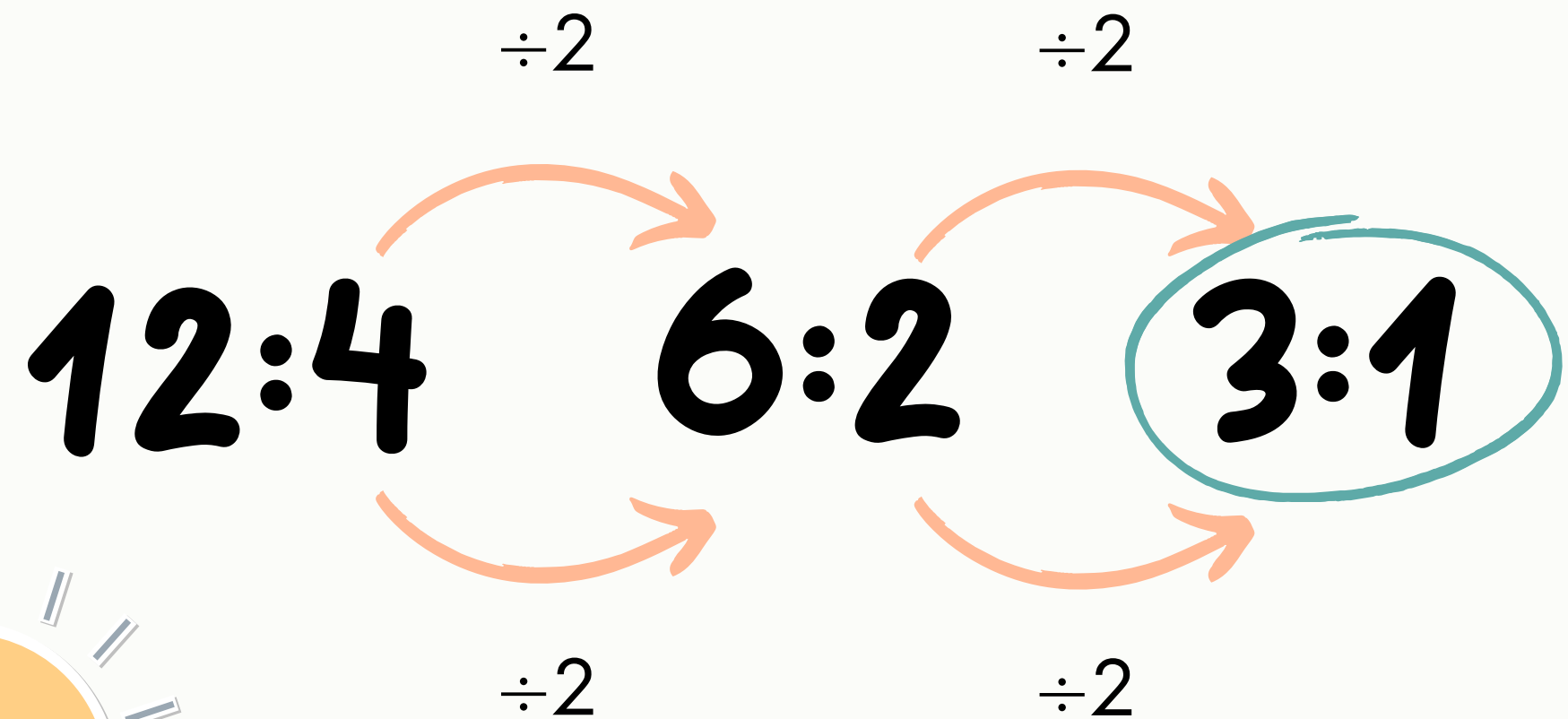
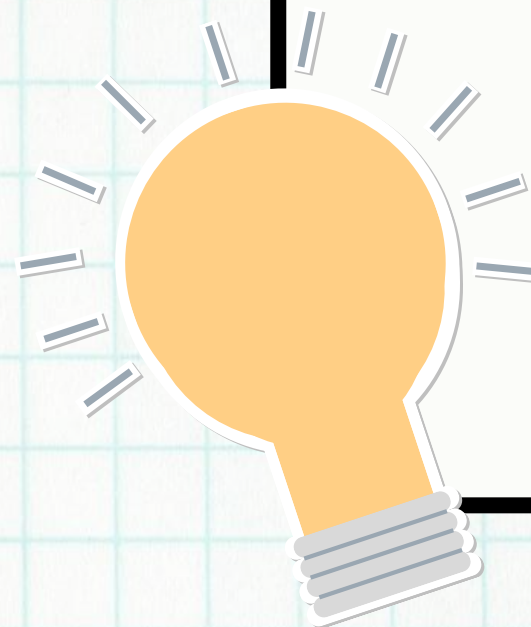


SOLVING RATIOS

3

Simplify, if possible.

Write the ratio of
blue crayons to red
crayons in simplest
form.



REAL LIFE APPLICATIONS

1

Paint mixing

To make lavender colored paint, mix 1 part violet to three parts white.



2

Cooking

My mom's famous cookie recipe has a two to one ratio of flour to sugar.

3

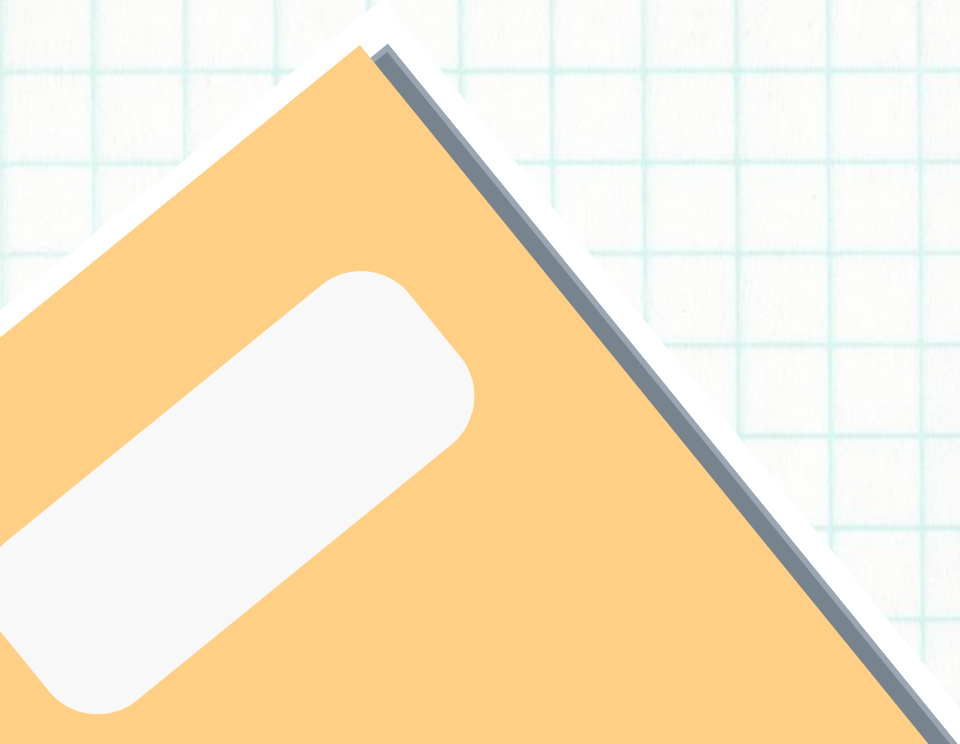
Scaled drawings

For every one inch on the drawing there is ten feet in real life.





**CAN YOU THINK OF ANOTHER
EXAMPLE OF HOW YOU MIGHT
USE RATIOS IN YOUR LIFE?**



The ratio of boys to girls in a class is 5:7. If there are 48 students in total, how many boys and how many girls are there?

A car travels 180km in 3 hours. How far will it go in 7 hours at the same speed?(include a model)

A recipe needs ingredients in the ratio 2:5:3(milk : flour : sugar). If you use 600 grams of flour, how much milk and sugar do you need?

A printer prints 90 pages in 6 minutes. How many pages per minute?

Which ratio is greater: **7:9** or **4:5**?

If 3 US dollars = 4 Canadian dollars, how many Canadian dollars in 27 US dollars?(include model)

RESOURCES

